

Un-requested Engine Start

Symptoms

- The engine cranks without the operator pushing the start button on the engine room panel or remote panel.

How To Use This Tree

This symptom tree can be used to troubleshoot un-requested engine start symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform depending on the symptom.

Shoptalk

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check Customer Interface Box	
	STEP 1A. Check Customer Interface Box Logic Unit	Is crank lamp illuminated?
STEP 2.	Check Engine Room Panel	
	STEP 2A. Check Engine Room Panel Control Panel	Less than 10 ohms resistance?
STEP 3.	Check Remote Panel	
	STEP 3A. Check Remote Panel Control Panel	Less than 10 ohms resistance?
STEP 4.	Check Customer Interface Box Logic Unit	
	STEP 4A. Check Customer Interface Box Logic Unit Voltage	+24 VDC?

STEP 1. Check Customer Interface Box

STEP 1A. Check Customer Interface Box Logic Unit

Conditions :

- Locate customer interface box
- Open customer interface box.

Action	Specification/Repair	Next Step
Verify crank lamp is illuminated.	Is crank lamp illuminated? YES	2A
	Is crank lamp illuminated? NO	4A

STEP 2. Check Engine Room Panel

STEP 2A. Check Engine Room Panel Control Panel

Conditions :

- Locate engine room panel
- Open engine room panel door
- Disconnect control panel connector X4.

Action	Specification/Repair	Next Step
Check the engine room panel control panel. • Place one test lead in the engine room panel start supply pin on the control panel X4 connector. • Place the other test lead in the engine room power switch supply pin on the control panel X4 connector.	Less than 10 ohms resistance? YES Repair : Replace the engine room panel control panel. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html).	Repair complete.
	Less than 10 ohms resistance? NO	3A

STEP 3. Check Remote Panel

STEP 3A. Check Remote Panel Control Panel

Conditions :

- Locate remote panel
- Open remote panel
- Disconnect remote panel control panel connector X4.

Action	Specification/Repair	Next Step
Check the remote panel control panel. • Place one test lead in the remote panel power switch supply pin on the control panel X4 connector. • Place the other test lead on the remote panel start supply pin of the control panel X4 connector.	Less than 10 ohms resistance? YES Repair : Replace the remote control panel control panel. Refer to Procedure 015-025 (/qs3/pubsys2/xml/en/procedures/115/115-015-025.html).	Repair complete.
	Less than 10 ohms resistance? NO	4A

STEP 4. Check Customer Interface Box Logic Unit

STEP 4A. Check Customer Interface Box Logic Unit Voltage

Conditions :

- Disconnect cable connector C1 from the customer interface box
- Open customer interface box.

Action	Specification/Repair	Next Step
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Check customer interface box logic unit voltage. - Place the positive test lead in the starter relay switch signal pin of the C1 connector. - Place the negative test lead in the starter relay return pin of the C1 connector.	24 VDC? YES Repair : Replace the customer interface box logic unit. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.
	24 VDC? NO Repair : Contact a Cummins Authorized Repair Location.	Repair complete.